CHAPTER 12 · CONCEPT 2

Population, Community and Interactions

Individuals make populations, populations make communities, and everything interacts.

Levels of organisation in nature

Term	Meaning
Population	A group of the **same kind** of organism in a habitat at a given time (e.g. all the rohu fish in a pond)
Community	**All the different populations** (plants, animals, microbes) sharing the same habitat

We can **estimate a population** by counting organisms in a marked area (e.g. a 1 m × 1 m plot).

Two kinds of interaction

- **Biotic–biotic** — among living things (a frog eats insects; a snake eats fish; plants compete for light).
- **Biotic–abiotic** — between living and non-living (earthworms live in moist soil; fish take oxygen from water).

Pollination is a key biotic interaction: insects, birds, wind or water carry pollen from stamens to carpels, helping plants make fruits and seeds.

Why variety matters: if a habitat had only one kind of organism, all would need the same resources, leading to competition and scarcity.

Where students lose marks

Trap 1 — Population = one kind; community = many kinds. A population is one species; a community is all the species together.

Trap 2 — Biotic–abiotic interaction involves a non-living factor. "Fish takes oxygen from water" is biotic–abiotic; "snake eats fish" is biotic–biotic.

Trap 3 — Pollination needs a carrier (insect, bird, wind, water) — it moves pollen so seeds/fruits can form.

Model answers — write them like this

Example A. 20 lotus plants grow in a pond. What does "20 lotus plants" represent?

Step 1: They are the same kind of organism in one habitat.

∴ A population of lotus plants.

Example B. Classify the interaction: "earthworms live in moist soil".

Step 1: Earthworm = biotic; moist soil = abiotic.

∴ A biotic–abiotic interaction.

Ready to practise? → Open the worksheet